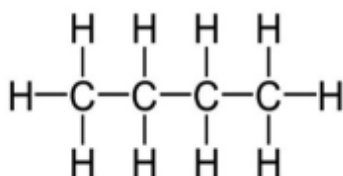


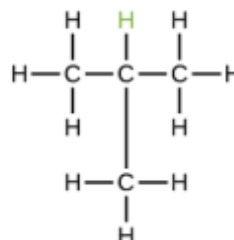
Isomers

Structural Isomer

Chain: Straight / Branch

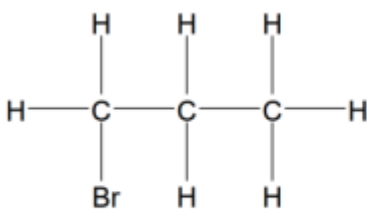


butane

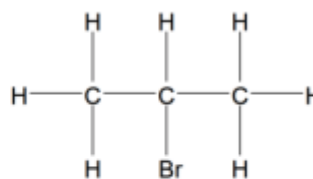


methylpropane

Positional: Functional Group is in different position

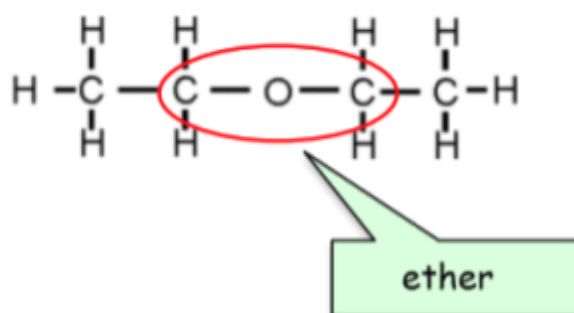
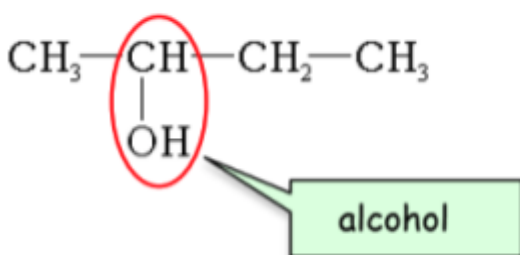


1-bromopropane



2-bromopropane

Functional Group: Different functional groups (eg: alcohol and ether)



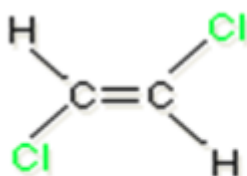
Geometric Isomer

"These are two different groups attached to each carbon atom on either side of the double bond. There is no rotation around a double bond"

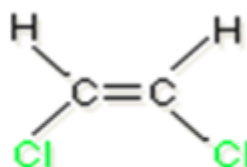
Cis-Trans Notation

Cis - Adjacent

Trans - Across



trans-1,2-dichloroethene



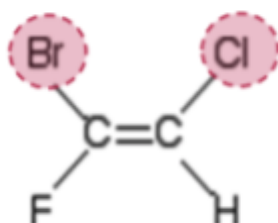
cis-1,2-dichloroethene

E-Z Notation

Higher atomic number higher priority, look at highest atomic number when it is same

Z - (Z)ame Side

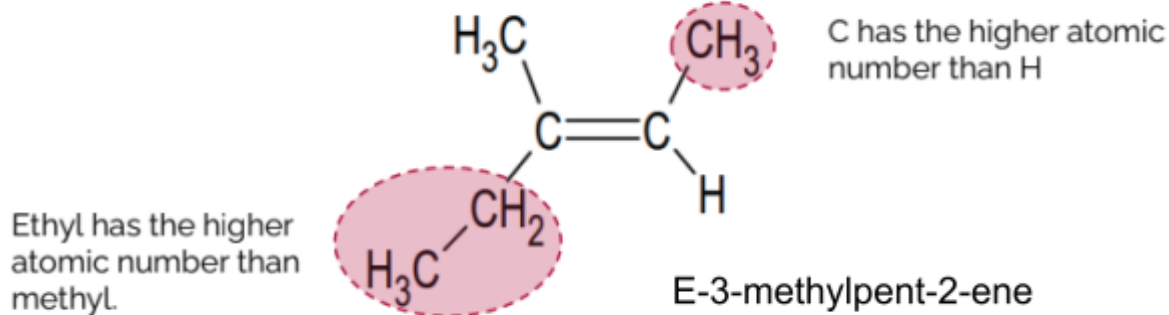
E - Opposite Side



Z 1-bromo-2-chloro-1-fluoroethene



E 1-bromo-2-chloro-1-fluoroethene



E-3-methylpent-2-ene

Alkane

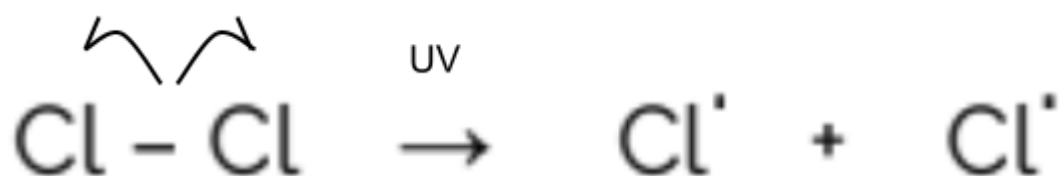
Free Radical Substitution

Initiation

Condition

- UV Light
- Vapour Phase Reaction

Homolytic Fission



Propagation

A radical attacking alkane forming another radical



Termination

Two free radicals collide which brings to an end of reaction.



Alkene

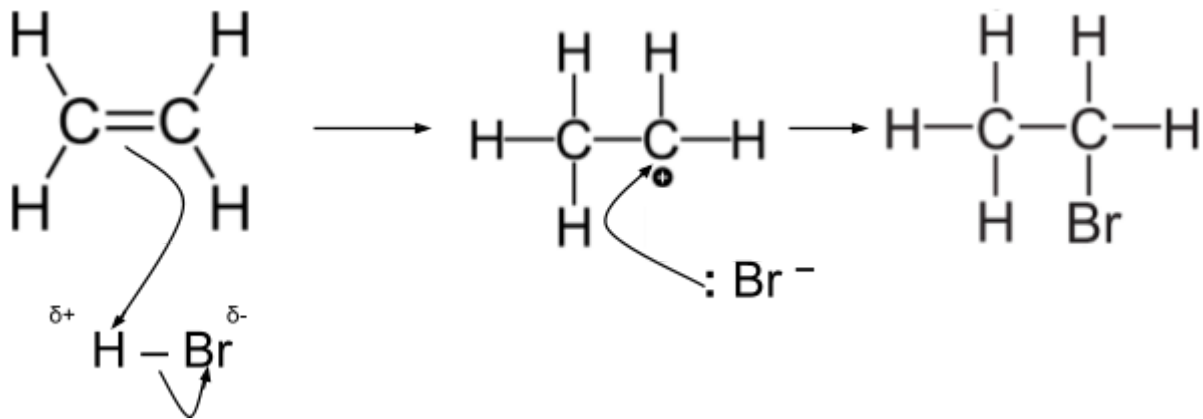
Electrophilic Addition

Electrophile Example

- Br_2
- H-Br

Condition

- Room temperature
- Bromine in a nonpolar solvent



Hydrogenation

Adds H_2 to the C=C double bond

Condition

- Catalyst: Nickel
- Temperature: 100°C
- Pressure: 3-5 atm

Margarine

Hydrogenation of fats

Halogenoalkane

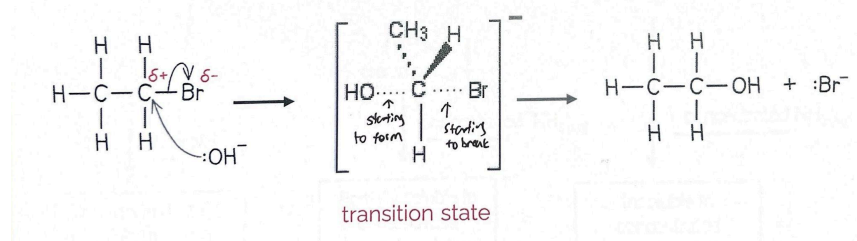
(S_N2) Nucleophilic Substitution

Nucleophiles: :OH⁻ (most common) :NH₃ :CN⁻

Reaction with NaOH

Conditions

- Aqueous NaOH solution
- Reflux
- Propanone as a mutual solvent



Reaction with KCN

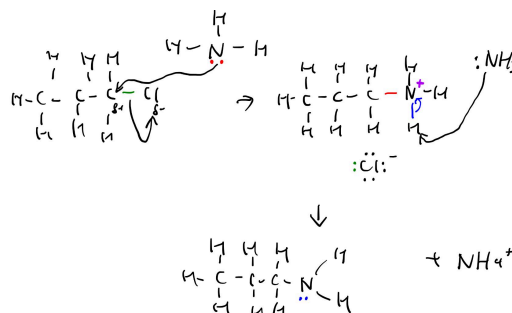
Conditions

- Alcoholic KCN solution
- Reflux

Reaction with Ammonia

Conditions

- Concentrated alcoholic ammonia solution
- Heated under pressure in a sealed tube

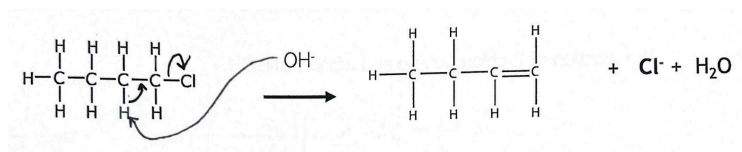


(E₂) Elimination Reaction

Conditions

- **NaOH in ethanol solvent**
- Reflux (Heat)

OH⁻ ion act as a base not a nucleophile



The C adjacent to the C-X must have a H attached for the H-X to be eliminated

Alcohol

Preparation of Alcohols

Direct Hydration (From Alkene to Alcohol)

Condition

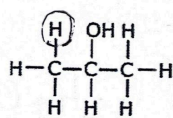
- Phosphoric acid on silica gel catalyst
- 300°C
- 60-70 atm

Dehydration Alcohol to Alkenes (Elimination)

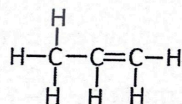
Condition

- Concentrated H₂SO₄ (dehydration agent + catalyst)
- Heat to 180°C

There must be an H on a carbon atom adjacent the carbon with the OH



propan-2-ol



propene

Manufacture of Ethanol from Ethene by Fermentation (Batch Process)

- No air (anaerobic condition)
- Kept under body temperature (25-40°C)
- Catalyst: Yeast / Zymase



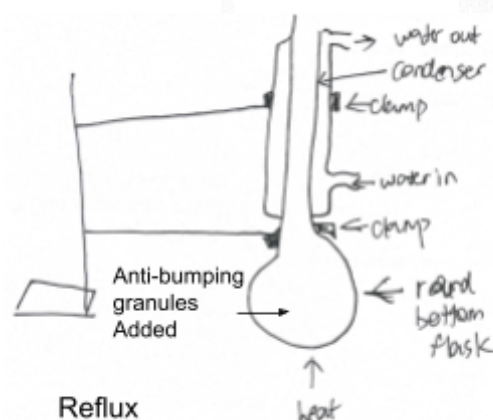
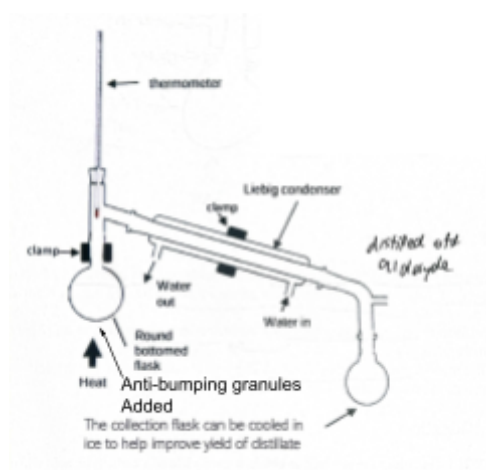
Oxidation of Alcohols

Condition

- Acidified Potassium Dichromate
- Concentrated Sulphuric Acid (Dehydrating Agent)

Primary Alcohol

1. Partial Oxidation
 - Distill out the aldehyde as it forms to prevent further oxidation to carboxylic acid
 - Excess alcohol
2. Full oxidation
 - Reflux
 - Excess oxidising agent to ensure maximum yield of carboxylic acid



Secondary Alcohol

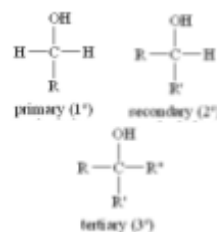
Condition

- Potassium Dichromate
- Conc Sulphuric Acid
- Reflux

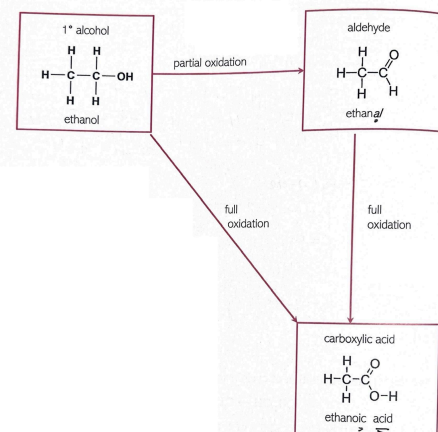
Tertiary Alcohol

NO Reaction

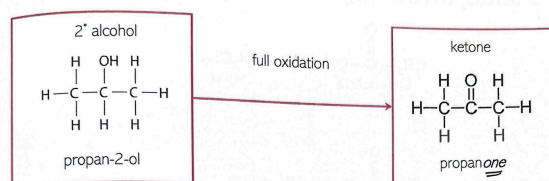
As it would require breaking of C-C bond



Primary alcohols

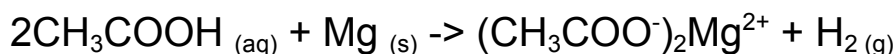


Secondary alcohols

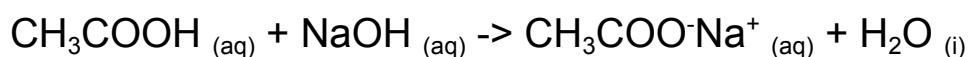


Reaction of Carboxylic Acid

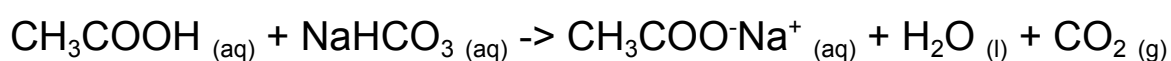
With Magnesium



With Sodium Hydroxide



With Carbonates and Hydrogencarbonates



Conversion to Esters (Carboxylic Acid $\rightarrow$ Esters)

Conditions

- **H₂SO₄ catalyst** (dehydrating agent = removing water)
- Reflux
- Anhydrous ZnCl₂ (absorbs water and push the position of equilibrium to right)



Naming: [*Alcohol*]+**yl** [*Carboxylic Acid*]+**oate**

Summary

Alkanes

- Free Radical Substitution

Alkenes

- Electrophilic Substitution
- Hydrogenation

Halogenoalkane

- Nucleophilic Substitution
- Elimination

Alcohol

- Direct hydration
- Fermentation
- Dehydration
- Oxidation
- Esterification
- Elimination